



NUS
National University
of Singapore



Market Lens

An AI-driven multi-agent system that turns scattered financial data into simple, confident investment signals (BUY, SELL, HOLD), backed by charts, sentiment and risk insights

IRS PM Project Report

Group 1

Jin YanYu (Team Leader)	A0326918U jinyanyu@u.nus.edu
Chen ZiHao	A0329043J
Fu HaoXiang	A0328896E
Li Lin	A0327882R
Yang Miao	A0327007M

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1 Introduction

1.1 Brief Overview

Market Lens is a multi-agent financial intelligence system that transforms fragmented market data into clear and evidence-backed investment insights. By emulating the workflow of a professional research team, the system distributes analytical responsibilities among specialized agents covering market trends, news sentiment, fundamentals, and social dynamics. Then their outputs are integrated through a manager-orchestrated reasoning loop. The final product is a Decision Card summarizing actionable investment signals (buy, sell, hold) with supporting charts, confidence levels, and quantified risk metrics. Built as a browser-based web application, Market Lens bridges the gap between advanced institutional analytics and the needs of everyday investors seeking clarity, explainability and control.

1.2 Importance and Relevance

In today's markets, investors have too much information but not enough clarity. Prices, news, and opinions change every minute, making it hard to know what really matters. Most people jump between different websites and apps and try hard to understand what's going on. This often leads to emotional or rushed decisions.

Market Lens solves this problem by using AI agents to collect and organize information automatically. It doesn't just show data—it explains it and provides educated reliable insights. Users can see where each conclusion comes from and how risks are managed. The system also helps reduce the time and effort needed to research which gives users structured insights they can trust.

For the academic side, this project shows how multi-agent reasoning systems and AI can work together to support human decision-making. For the business side, it demonstrates how intelligent assistants can help retail investors think and act like professionals—faster, smarter, and with more confidence.

1.3 Main Goals and Objectives

The Main Goals of Market Lens are:

- (1) Build an AI agent system that works like a real investment team, where each agent has a clear role and communicates effectively with others.
- (2) Turn complex, unstructured data into clear and useful investment insights using Decision Cards.

Main Objectives include:

- Developing a stable data pipeline that gathers and cleans information from multiple sources.
- Designing clear reasoning and debate between bullish and bearish views to reach balanced conclusions.
- Adding risk controls for different risk levels (aggressive, neutral, conservative).
- Building an easy-to-use web interface where users can chat with the system and adjust analysis settings.

2 Project Background and Market Context

2.1 Background and Motivation

In today’s fast-moving markets, investors have access to more information than ever before—stock prices, company reports, financial news, and even social media opinions. Yet, having more data does not always mean better decisions. Retail investors often face a constant flow of updates across multiple platforms, forcing them to manually filter, interpret, and compare different sources before acting. This process is time-consuming and mentally exhausting, especially for those who are not full-time traders.

Market Lens was created to solve this challenge. Our goal is to make investment research simpler, clearer, and faster by using AI agents to automate information gathering and reasoning. Instead of reading through endless reports or headlines, users can receive structured insights within minutes. The final decision cards summarize whether to buy, sell, or hold a stock, backed by key evidence and reasoning.

2.2 The Challenge

Most existing tools focus on showing data, not explaining it. Free platforms like [Yahoo Finance](#) and [Google Finance](#) provide large amounts of raw information but leave users to interpret it on their own. Professional terminals like [Bloomberg](#) offer deep analysis but are expensive and complex. On the other end, social media channels such as Telegram or [Reddit](#) make market information more accessible but often unreliable, filled with hype or unverified opinions. (More details can be found in Section 4. Market Research)

This creates a clear gap: investors need a tool that combines automation, clarity, and trust—something that filters noise, highlights what matters, and presents it in a form they can easily understand and quickly act on.

2.3 Overview of Market and Academic Landscape

The finance industry is gradually adopting AI to process data and detect trends, but most solutions still specialize in narrow areas. Some focus on sentiment analysis from news and tweets, while others handle quantitative price forecasting. Very few systems combine multiple perspectives which are news, market signals, fundamentals, and risk factors into a single, interpretable framework.

From an academic view, this project connects research in multi-agent reasoning systems and explainable AI (XAI) with a real-world finance application. It demonstrates how multiple AI agents can collaborate, debate, and reason to produce transparent decisions, aligning closely

with topics taught in Machine Reasoning (MR), Reasoning Systems (RS), and Cognitive Systems (CGS) courses.

2.4 Significance of the Challenge

Information overload is now one of the biggest obstacles for both new and experienced investors. When decisions are made under pressure or based on incomplete understanding, it often leads to poor timing, unnecessary risk, and emotional trading.

Market Lens directly addresses these pain points by:

- Reducing cognitive load through automated data gathering and structured reasoning.
- Increasing transparency with clear evidence for every decision.
- Offering fast, consistent insights that keep users informed without being overwhelmed.

This project sits at the intersection of AI innovation and financial accessibility. It captures a growing opportunity: helping investors transform information overload into insight—making the market a little less noisy and a lot more understandable. In summary, while the academic and technological foundations for AI-assisted reasoning are rapidly evolving, their commercial application to retail investing remains limited. Most current products either emphasize algorithmic prediction without transparency or focus on data presentation without synthesis. This gap between research capability and real-world accessibility defines the opportunity space for Market Lens. The following section presents the business rationale behind the system: analysing market demand, competitive landscape, user needs, and strategic positioning.

3 Market Research

The purpose of this section is to evaluate the commercial viability and strategic positioning of Market Lens within the current investment technology landscape. Building upon the challenges identified in Section 2, this analysis examines market demand, competitive dynamics, user needs, and potential differentiation. It integrates both academic insights from explainable AI and practical findings from industry research to demonstrate how Market Lens addresses a clear market gap between data accessibility and decision intelligence. By combining structured reasoning, transparency, and affordability. The system aims to translate advanced AI methods into a sustainable business opportunity with measurable impact.

3.1 Market Analysis: Demand, Trends and Opportunities

The global retail trading market continues to expand, driven by the rise of digital platforms and a new generation of investors. A growing number of individuals now trade directly through mobile apps, seeking autonomy and faster access to information. This shift has created a strong demand for tools that can simplify complex data and translate it into clear investment insights. However, as data sources multiply—covering news, financial reports, and social sentiment—investors are increasingly overwhelmed. Many rely on surface-level signals or social media opinions instead of structured analysis. This environment highlights a major opportunity for systems that can organize, interpret, and explain market information with transparency and confidence. Market Lens addresses this opportunity by introducing an AI-driven reasoning framework that mimics how professional research teams filter and reason through market data, yet remains lightweight and accessible to everyday users.

3.2 Key Competitors and Current Solutions

The current market for investment tools can be grouped into four main categories:

- (1) **Institutional Terminals Platforms** like [Bloomberg Terminal](#) and [LSEG](#) provide unmatched data depth, real-time analytics, and professional research tools. They remain the industry standard for institutional investors but are too expensive and complex for most individuals. Their design also assumes financial expertise, often leading to information overload for non-experts.
- (2) **Retail Information Platforms**, such as [Yahoo Finance](#) and [Google Finance](#) have made basic data—charts, earnings, and news—widely accessible. Yet, they do not synthesize the data. Users still need to interpret raw numbers and headlines themselves, which can lead to inconsistent or emotionally driven decisions.

- (3) **Social and Community Platforms** like [Robinhood](#) and [eToro](#) attract users through social trading, community feeds, and gamified design. While these features increase participation, they also encourage trend-following and reduce analytical depth, often prioritizing popularity over evidence-based reasoning.
- (4) As for **Emerging AI-Assisted Tools**, recent startups such as [Fiscal.AI](#), [Kavout](#), and [Quiver Quant](#) apply natural language processing and sentiment analysis to financial news or filings. These innovations represent progress toward intelligent automation but remain narrow in scope, often analysing only one signal type (e.g. sentiment only) without combining multiple perspectives like fundamentals, technical and risk evaluation.

3.3 User Needs and Potential Differentiation

User research and market observation reveal three recurring pain points among retail investors:

- (1) **Information overload:** too many unfiltered data sources create confusion instead of clarity.
- (2) **Lack of synthesis:** most platforms show data but do not connect it into actionable insights.
- (3) **Trust and transparency gaps:** many users follow social trends because they lack explainable guidance.

Market Lens differentiates itself by addressing these gaps directly. Instead of giving users more data, it gives them structured understanding. Through multiple AI agents working together—each focusing on one domain and reporting to a coordinating manager—the system produces clear, auditable Decision Cards that summarize the “what,” “why,” and “risk” behind each recommendation. This approach combines the rigor of institutional analysis with the simplicity and accessibility expected by modern retail investors.

In summary, while existing solutions either overwhelm or oversimplify, Market Lens fills the missing middle: a transparent, multi-agent reasoning platform that transforms information noise into focused, explainable investment intelligence.

4 Project Scope

4.1 Techniques Focus

The project focuses on several core aspects of intelligent reasoning systems:

- **Multi-Agent Reasoning Architecture:** Modelling division of labour, collaboration, and adversarial debate among LLM-based agents (Analyst, Researcher, Trader, Risk Manager).
- **Structured Decision-Making:** Generating interpretable and evidence-based decision cards through reasoning chains and confidence-weighted synthesis.
- **Closed-Loop Automation:** Integrating data collection, reasoning, risk control, and execution into a coherent, self-improving workflow.
- **Explainability and Transparency:** Ensuring each recommendation is traceable through reasoning chains, debate matrices, and risk annotations.

4.2 Limitations and Constraints

- **Data Constraints:** The system relies on public APIs (e.g., Yahoo Finance, Reddit, Finnhub), which may limit data granularity and historical depth compared to commercial datasets.
- **Academic Constraints:** This system is developed for educational and experimental purposes under the NUS-ISS Intelligent Reasoning Systems module. While it demonstrates strong conceptual validity and practical potential, it is not intended for real-world financial trading, regulatory compliance, or live deployment.
- **Time and Resource Limitations:** As a semester-long project, development focuses on functional demonstration and reasoning explainability rather than full-scale scalability or optimization.

5 Data Collection & Preparation

Market Lens combines diverse financial data streams with natural-language knowledge extraction to power intelligent reasoning and coordinated agent workflows. Data is sourced from open APIs, automated scraping pipelines, and professional market data providers to ensure wide coverage and balanced quality. The sections below outline the system’s data inputs, collection pipeline and data-processing architecture.

5.1 Data Sources

(1) Market Data

- Providers: Yahoo Finance API (via [yfinance](#) library)
- Data Types: Real-time stock prices (open, close, high, low, volume), fundamental indicators (market capitalization, P/E ratio, P/B ratio, dividend yield, Beta), and historical K-line data (timeframes from 1 minute to 1 month).
- The system computes fifteen commonly used technical indicators—such as Moving Average (MA), Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), Bollinger Bands (BB), Stochastic Oscillator (SO), and Average True Range (ATR)—as structured quantitative features that support the agents’ analytical reasoning.

(2) News Data

- Providers: Finnhub API, Financial Modeling Prep (FMP), and Alpha Vantage.
- Data Types: Aggregated global financial and corporate news, including earnings announcements, analyst opinions, and market reports.

(3) Fundamental Data

- Providers: FMP and Finnhub.
- Data Types: Company profile, key financial metrics, balance sheet and income statement data, as well as recent institutional transaction activities.

(4) Social Media Sentiment Data

- Acquisition: Automated crawling via *Playwright* headless browser.
- Collection Strategy: Multi-keyword search combining ticker symbols and financial sentiment keywords.

5.2 Data Acquisition and Processing Workflow

Market Lens employs an automated asynchronous pipeline implemented with *asyncio*, enabling efficient concurrent data retrieval from multiple APIs and web sources. All incoming data is stored in date-organized JSON format within structured databases, with

independent caching for each source. This design supports incremental updates and minimizes redundant API calls.

The **Market Lens** system employs a structured, multi-stage approach to ensure data consistency, reliability, and semantic interoperability across heterogeneous financial and textual sources. The key processes include:

- (1) **Cleaning and Normalization:** Raw market and textual data are subjected to duplicate removal, timestamp alignment, and ticker code standardization. These procedures eliminate redundancy and temporal misalignment while harmonizing identifiers across datasets.
- (2) **Text Processing (News and Social Data):** Natural Language Processing (NLP) models are applied to extract sentiment and thematic topics from unstructured text sources. An intelligent deduplication module further consolidates information based on title similarity and publisher identity. Multi-level quality scoring mechanisms—integrating engagement metrics (e.g., upvotes, comments) and textual richness—are used to quantify content reliability. A multi-layer filtering pipeline is subsequently employed to preserve only credible and contextually consistent information.
- (3) **Unified Data Representation:** All processed outputs are standardized into a **JSON-based schema**, enabling seamless integration with downstream reasoning agents and efficient storage within vector databases for semantic retrieval and similarity search.

5.3 Key Challenges

Despite the robustness of the Market Lens framework, several challenges were encountered during system development and experimentation:

- (1) **Data Continuity and Redundancy:** Ensuring uninterrupted data availability required the implementation of local caching and multi-source redundancy mechanisms. These strategies mitigate downtime from API limits or transient network disruptions but introduce additional complexity in synchronization and version control.
- (2) **Limitations of Public Data Sources:** While sufficient for research, testing, and simulation, publicly available financial datasets do not always match the depth, granularity, or update frequency of commercial-grade databases. This constraint affects the system’s ability to perform ultra-low-latency analysis or incorporate premium market signals typically used in institutional environments.
- (3) **Variability and Bias in Textual Data:** Social media streams often exhibit high linguistic variability, sarcasm, and sentiment ambiguity, making consistent interpretation

challenging. Meanwhile, professional news sources can carry political, cultural, or editorial biases that influence the perceived sentiment of financial events. Addressing these factors necessitated multi-layer filtering and scoring mechanisms to enhance data reliability and neutrality.

6 System Design

6.1 Architecture and Components

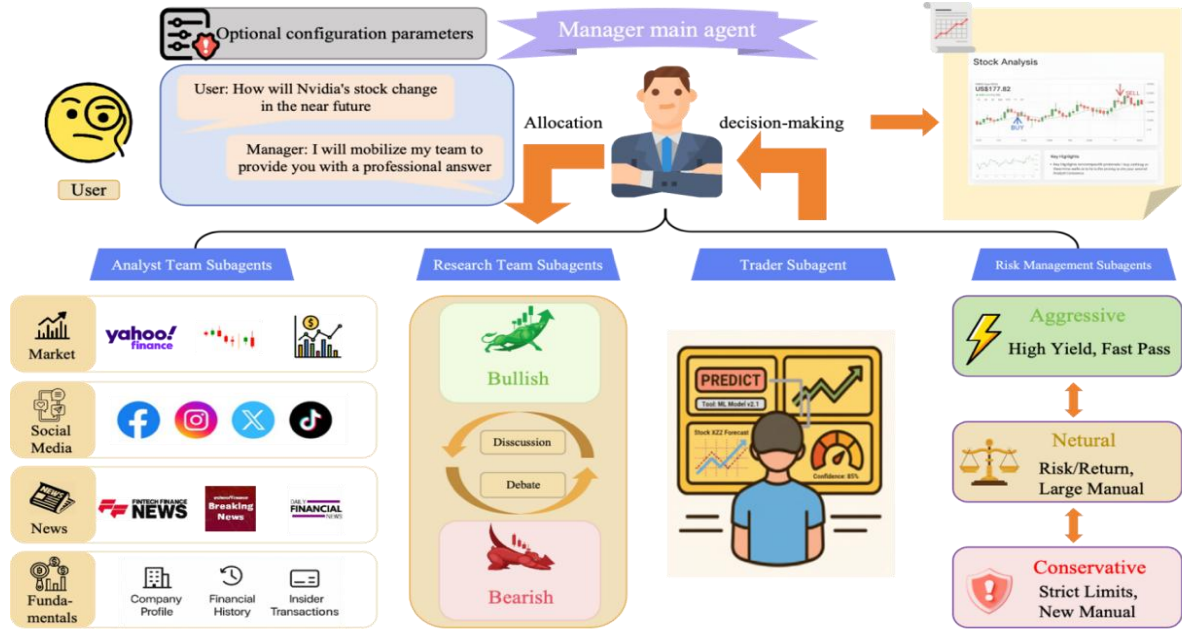


Figure 1 System Architecture

Market Lens adopts a multi-agent, closed-loop architecture designed to satisfy both academic objectives (traceable reasoning, explainable AI) and market requirements (real-time responsiveness, transparency, and risk control).

At the top level, a React single-page application (SPA) provides an interactive interface for communication, chart visualization, and decision card rendering. It connects to the backend through a FastAPI gateway using both WebSocket (for real-time streaming of reasoning logs and incremental updates) and REST APIs (for configuration, file upload, and session management).

The FastAPI backend manages authentication, routing, and message streaming. It forwards user requests to the Manager Agent, which interprets intent, constructs an execution plan, orchestrates downstream agents, validates results via JSON schema, and queues tasks using asynchronous scheduling (with optional Celery/Redis integration).

The cognitive reasoning layer includes four specialized agents working in coordination:

- **Analyst (LangChain toolset):** Performs asynchronous multi-source data gathering and preprocessing (market data, news, fundamentals, and social sentiment). It computes technical indicators, extracts NLP-based sentiment and topics, and outputs normalized JSON datasets with caching and time window alignment.

- **Researcher:** Executes multi-round debates between Bullish and Bearish agents, moderated by a Moderator agent that synthesizes balanced, evidence-backed conclusions (BUY/SELL/HOLD) with confidence scores.
- **Trader:** Converts conclusions into Decision Cards detailing entry/exit ranges, stop-loss/target levels, and position sizing. It can optionally integrate the Kronos deep forecasting model to generate short-term quantitative predictions.
- **Risk Management:** Evaluates each Decision Card under Aggressive, Neutral, and Conservative profiles, automatically adjusting exposure and thresholds. It produces a unified risk report accompanied by human-readable annotations.

The data plane leverages public APIs, local caches, snapshots, and vector databases to maintain balance between freshness, reproducibility, and system efficiency. Results are streamed to the React frontend in real time via WebSocket, ensuring responsive visualization and smooth user experience.

6.2 Reasoning Technique Selection

Evaluated Approaches:

- **Single-Agent Chain-of-Thought (CoT):** Simple but lacks robustness against conflicting evidence and limited traceability.
- **Tool-augmented reasoning (ReAct / Tree-of-Thought):** Effective for retrieval and procedural reasoning, yet weak in multi-perspective synthesis.
- **Multi-Agent Debate Architecture:** Enables adversarial reasoning, promotes transparency, and supports evidence-based consensus.
- **Pure Quantitative Forecasting:** Strong predictive power but low interpretability; requires explainable outer reasoning layer.

Final Approach:

Market Lens adopts a **hybrid multi-agent debate framework** for cognitive robustness and explainability.

- The **Analyst** grounds reasoning in verifiable data.
- The **Researcher** layer embodies structured argumentation and moderation.
- The **Trader** fuses symbolic reasoning with quantitative forecasting through Kronos.
- The **Risk Management** module enforces risk-aware consistency.

6.3 Module Interaction Overview

- **Manager** → **Analyst:** Task plan → data collection → structured file

- **Manager** → **Researcher:** Facts → multi-round debate → conclusion + confidence + evidence
- **Manager** → **Trader:** Research output → Decision Card (+ optional Kronos forecast)
- **Manager** → **Risk:** Decision Card → adjusted strategy + unified risk report
- **FastAPI** ↔ **React:** Real-time WebSocket data stream + REST-based session/config management

7 Implementation

The Market Lens system has now achieved full integration across all core modules—Manager, Analyst, Researcher, Trader, and Risk Management—forming a closed-loop reasoning pipeline capable of generating interpretable trading insights from real financial data.

This section summarizes the practical workflow and implementation progress without duplicating result analysis.

7.1 Workflow Overview

Users are first required to **log in**, after which the system automatically stores all relevant user information for personalized interactions. Alternatively, users may select **Guest Access** to quickly explore and experience the platform without creating an account.

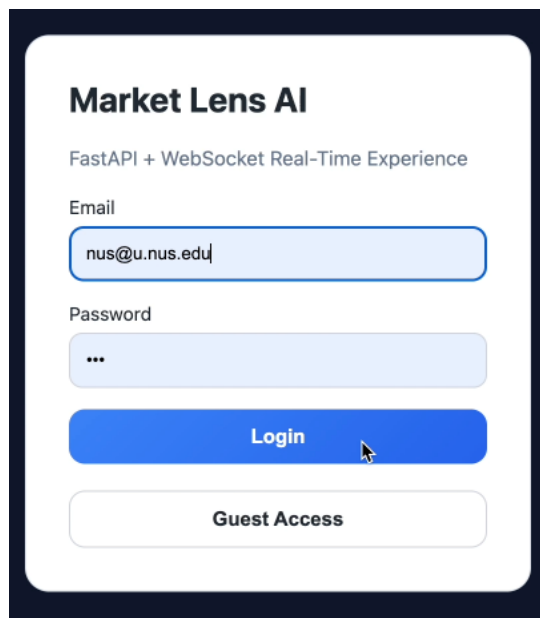


Figure 2 Login Page

The complete user interaction begins in the **React-based frontend**, where users communicate with the **Manager Agent** through natural language queries via a **WebSocket connection** managed by the **FastAPI backend**. At the same time, users can **configure the analysis parameters and content** according to their specific requirements.

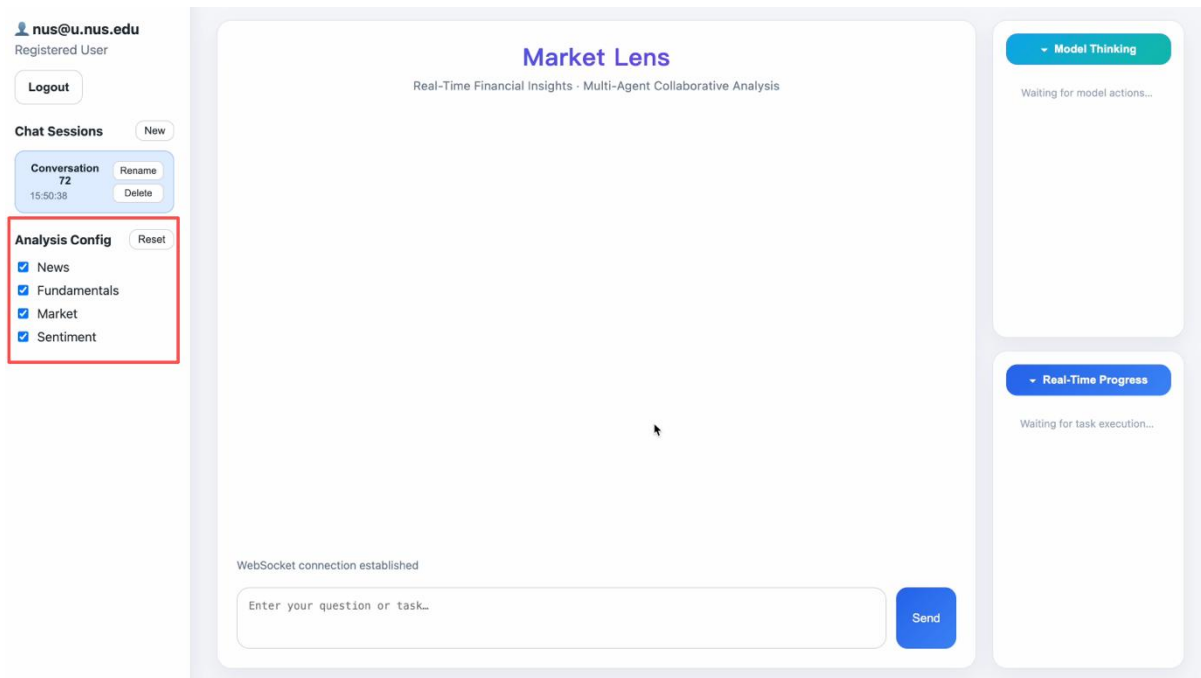


Figure 3 Home Page

The **Manager** interprets each request (e.g., “analyze Apple stock for next week”) and **adaptively invokes the appropriate sub-agents** from the following modules to accomplish the task:

```
# Dynamic configuration description
if config:
    enabled_types = [k for k, v in config.items() if v]
    analysis_desc = f"Currently enabled: {'', '.join(enabled_types)}" if enabled_types else "All analysis types disabled"
else:
    analysis_desc = "All analysis types enabled"

prompt = ChatPromptTemplate.from_messages([
    ("system",
     "You are the Market Lens AI main agent, a professional, intelligent, and autonomous financial market analysis assistant.\n\n"
     "🎯 Core Mission: Quickly understand user intent, rationally coordinate tool chains, and generate actionable market insights and trading recommendations.\n\n"
     "🛠️ Available Tools:\n"
     "1. call_analyst: Collect stock raw data (news / fundamentals / market / sentiment).\n"
     "2. call_researcher: Conduct in-depth research based on data (bull/bear debate, investment recommendations).\n"
     "3. call_trader: Generate trading decisions, supports user_request parameter to pass user needs.\n"
     "4. call_risk_manager: Review or supplement risk conclusions when necessary.\n"
     "5. read_file / write_file: Manage and track historical files, notes, and intermediate results.\n\n"
     "🔄 Autonomous Management:\n"
     "- Before each response, determine if existing conversation and saved files can directly meet needs; first retrieve relevant summaries or historical results via read_file.\n"
     "- When obtaining new conclusions, key parameters, or action items, use write_file to save concise notes for future reference; confirm before writing to avoid overwriting content that needs to be retained.\n"
     "- Avoid repeatedly running expensive tools. Unless needs change or historical files are missing/outdated, prioritize reusing existing results and explain the basis for reuse in the response.\n"
     "- Tool calls must align with user goals: collect data when lacking, research deeply when logic exists, execute when strategy is present, and verify when risks exist.\n\n"
     "📋 Recommended Workflow:\n"
     "- Complete research: call_analyst - call_researcher - call_trader - (if necessary) call_risk_manager.\n"
     "- In-depth research: call_analyst = call_researcher.\n"
     "- Quick query or reuse history: read_file to retrieve notes - call relevant tools if updates needed.\n\n"
     "⚠️ Response Requirements:\n"
     "- Do not reveal any actual file paths or internal storage locations in user responses.\n"
     "- When tools return human_readout, make it the core of the response and supplement with necessary context.\n"
     "- Output must be specific: include direction, position, stop-loss/take-profit, trigger conditions and other key values; explain gaps if data is insufficient.\n"
     "- When risk control fails, remind users to proceed with caution and provide current trading points.\n\n"
     "📊 Configuration Status: {analysis_desc}\n\n"
     "📝 Key Notes:\n"
     "- Keep JSON as-is for passthrough, do not arbitrarily modify fields.\n"
     "- When calling call_trader, always write user's original request into user_request parameter.\n"
     "- If user mentions prediction, price outlook, or Kronos, explicitly record in user_request.\n"
     "- When partial analysis fails but usable data exists, continue and inform about missing parts and impact.\n"
     "- Respond to users in English throughout, maintaining a professional, clear, and actionable tone."),
    ("human", "{input}"),
    MessagesPlaceholder("agent_scratchpad")
])
```

Figure 4 System Prompt for Manager

(1) Analyst Phase – Data Aggregation and Analysis

The Manager triggers the Analyst module (analysts/analyst.py), which coordinates four analytical tools—news, fundamentals, market data, and sentiment—under a unified LangChain agent.

These tools retrieve and preprocess multi-source information in parallel, performing sentiment scoring, financial ratio extraction, and technical indicator computation.

The system stores all processed outputs as structured JSON, forming the analytical foundation for reasoning tasks.

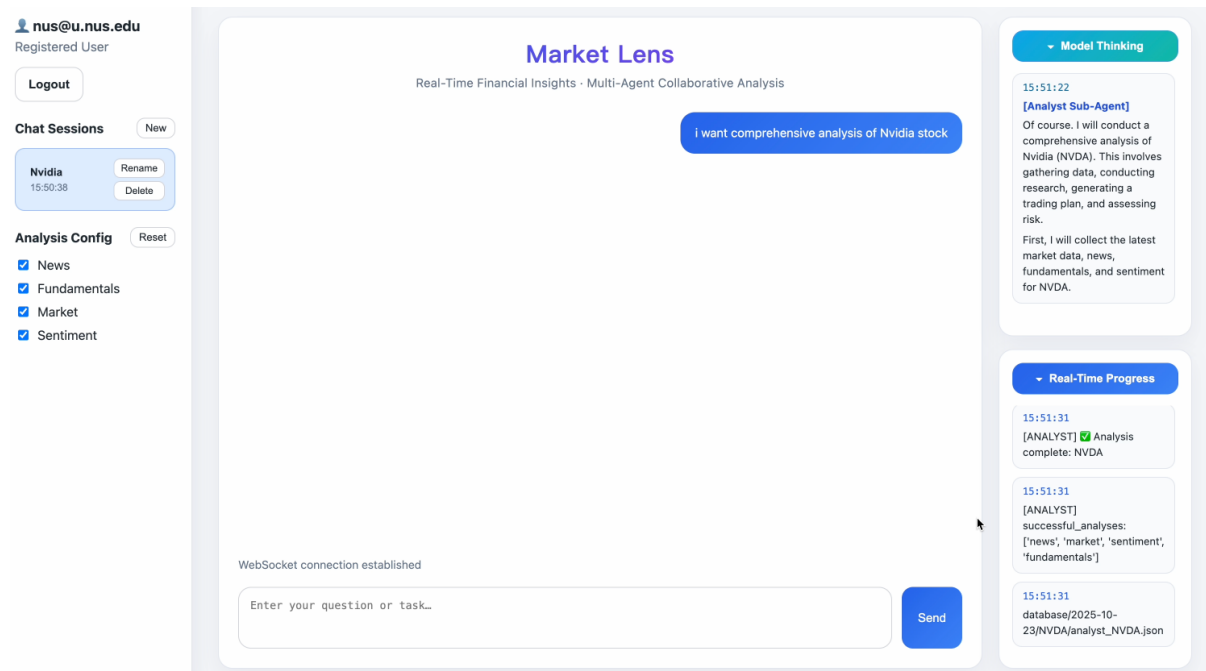


Figure 5 Communication with Manager

```
prompt = ChatPromptTemplate.from_messages([
    [
        SystemMessage(
            content=(
                "You are the Analyst, invoked by a Main Manager.\n"
                "You will receive a single stock ticker and an analysis intent.\n\n"
                "Available tools:\n"
                "1) get_news → latest headlines/summaries about a ticker or company.\n"
                "2) get_fundamentals → company basics, key financial metrics, insiders.\n"
                "3) get_market → market/price/quote/volume/trading-session context.\n"
                "4) get_sentiment → overall sentiment/stance beyond single-article headlines.\n\n"
                "Routing rules:\n"
                "- Choose exactly one tool unless the request clearly asks for multiple.\n"
                "- If multiple are requested, call them sequentially then summarize jointly.\n"
                "- Prefer get_news for specific events; get_sentiment for broader mood.\n\n"
                "OUTPUT FORMAT REQUIREMENT:\n"
                "Return JSON ONLY with keys exactly:\n"
                "  ticker: the input ticker (string)\n"
                "  channel: one of ['news','fundamentals','market','sentiment']\n"
                "  data: the raw JSON returned by the tool you called\n"
                "  summary: a short, clear 1-3 line summary for the Manager\n"
            )
        ),
        MessagesPlaceholder(variable_name="history"),
        ("human", "{input}"),
        MessagesPlaceholder(variable_name="agent_scratchpad"),
    ]
])
```

Figure 6 System Prompt for Analyst

(2) Researcher Phase – Debate and Synthesis

The Manager then invokes the Researcher Manager (researchers/manager.py), which validates the Analyst data and initiates multi-round debates between **bullish** and **bearish** reasoning tools. Each agent presents arguments, evidence, and uncertainties; a moderator consolidates their discussion into a balanced conclusion with a clear BUY/SELL/HOLD stance and quantified confidence score.

This phase embodies adversarial reasoning, ensuring transparency and interpretability.

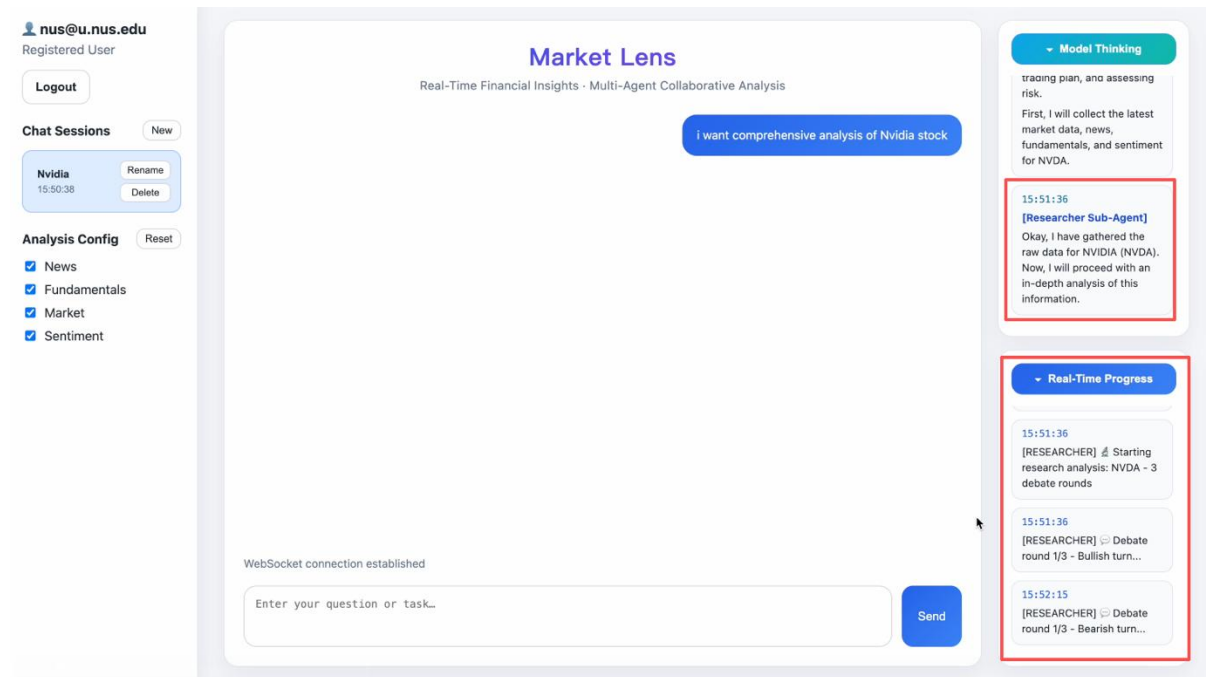


Figure 7 Model Thinking & Real-Time Progress

```
PROMPT = ChatPromptTemplate.from_messages([
    ("system",
        "You are the Market Lens debate moderator, consolidating bullish and bearish briefs into a decision memo for the investment committee.\n"
        "Operating standards:\n"
        "- Remain neutral and audit-ready; reconcile conflicts with explicit references to provided evidence only.\n"
        "- Weight arguments by evidence quality and freshness; flag unsupported claims rather than amplifying them.\n"
        "- Ensure the action plan respects the stated risk_tolerance and time_horizon.\n"
        "- Deliver strict JSON that conforms to the schema; narrative fields must stay in English."),
    ("user",
        "Ticker: {ticker}\n"
        "Risk tolerance: {risk_tolerance}\n"
        "Default horizon: {time_horizon}\n\n"
        "Bullish JSON:\n{bull_json}\n\n"
        "Bearish JSON:\n{bear_json}\n\n"
        "JSON Schema (target output):\n{schema}\n\n"
        "Instructions:\n"
        "1) Summarize both theses.\n"
        "2) Extract consensus facts (both sides implicitly/explicitly accept).\n"
        "3) List disagreements as {{topic, bull_view, bear_view}}.\n"
        "4) Distill key upside and key risks (use strongest, well-evidenced points).\n"
        "5) Build a scorecard: bull_strength, bear_strength (0-1), uncertainty (0-1), net_score = bull-bear-uncertainty.\n"
        "6) Map net_score to a recommendation (BUY if > +0.15, SELL if < -0.15, else HOLD); adjust by risk_tolerance and time_horizon.\n"
        "7) Provide triggers_up / triggers_down (objective milestones that would change the call).\n"
        "8) Include evidence_citations as a deduped list from both sides' evidence.\n"
        "Quality checks:\n"
        "- If evidence is insufficient for a field, include a concise explanation while keeping the schema intact.\n"
        "- Return ONLY valid JSON; no surrounding text.")
])
```

Figure 8 System Prompt for Researcher

(3) Trader Phase – Strategy Generation

The Trader Agent (Trader/trader.py) translates research outcomes into actionable strategies.

It can optionally activate the **Kronos deep prediction model**, which forecasts short-term price trends based on encoded OHLCV sequences.

The Trader then generates standardized Decision Cards including trade direction, entry/exit range, position size, and confidence level, ready for risk evaluation.

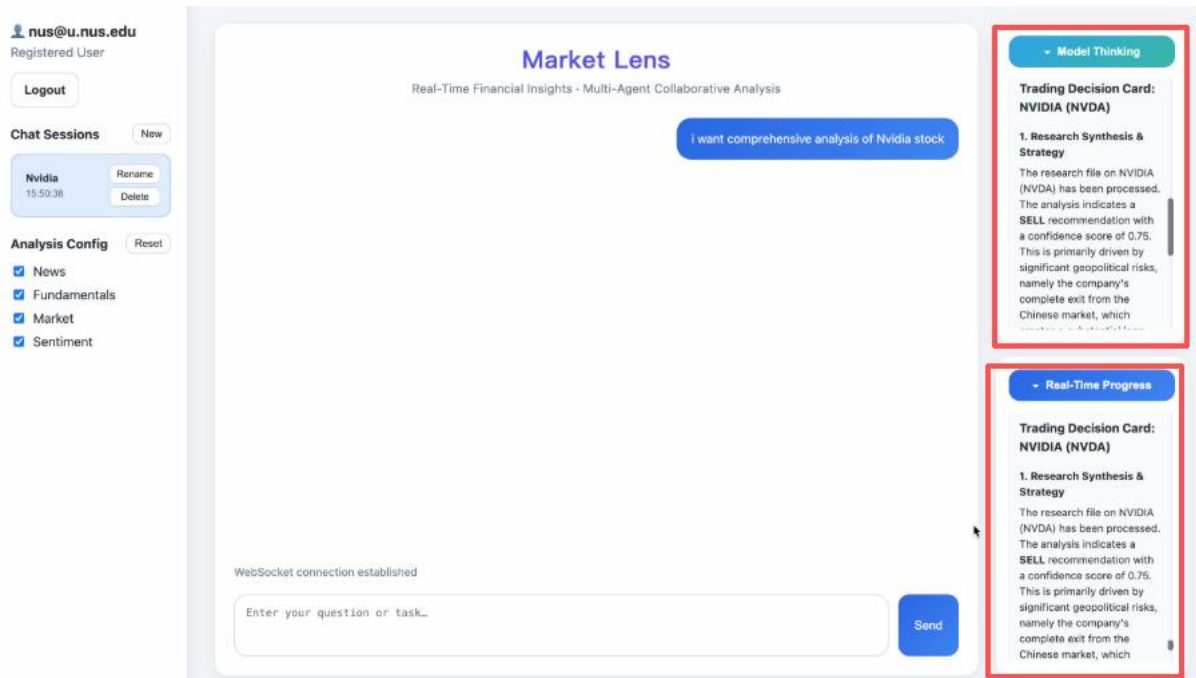


Figure 9 Trading Decision Card

```
prompt = ChatPromptTemplate.from_messages([
    ("system",
        "You are the Market Lens trading officer. Translate research outputs into actionable execution plans.\n"
        "Duties:\n"
        "- Fully digest the research JSON, align with user intent, and craft a coherent strategy.\n"
        "- Maintain a professional, auditable tone; never fabricate missing data-state assumptions explicitly.\n"
        "Available tools:\n"
        "1) load_research_data [ ] load research outputs (including csv_path fields).\n"
        "2) run_kronos_prediction [ ] invoke Kronos when additional forecasting is warranted.\n"
        "3) generate_decision_card [ ] produce a structured trading decision card.\n"
        "Operating guidelines:\n"
        "a. Call load_research_data first and inspect the symbols list.\n"
        "b. Based on user requests, research uncertainty, and csv_path availability, decide whether to run Kronos per symbol; when used, capture prediction_data in the decision card.\n"
        "c. Use generate_decision_card to synthesize research takeaways, risk controls, and sizing, incorporating Kronos insights when provided.\n"
        "Kronos rules of engagement:\n"
        "- Mandatory when the user explicitly asks for forecasts / price outlook / Kronos.\n"
        "- Recommended for "comprehensive analysis" requests or when uncertainty is high / very_high.\n"
        "- When csv_path is available and the user signals interest, run Kronos and reference the key outputs.\n"
        "The final answer must reconcile research, live predictions, and risk controls with clear sourcing.",
        MessagesPlaceholder("chat_history"),
        ("user", "{input}"),
        MessagesPlaceholder("agent_scratchpad"),
    ])

```

Figure 10 System Prompt for Trader

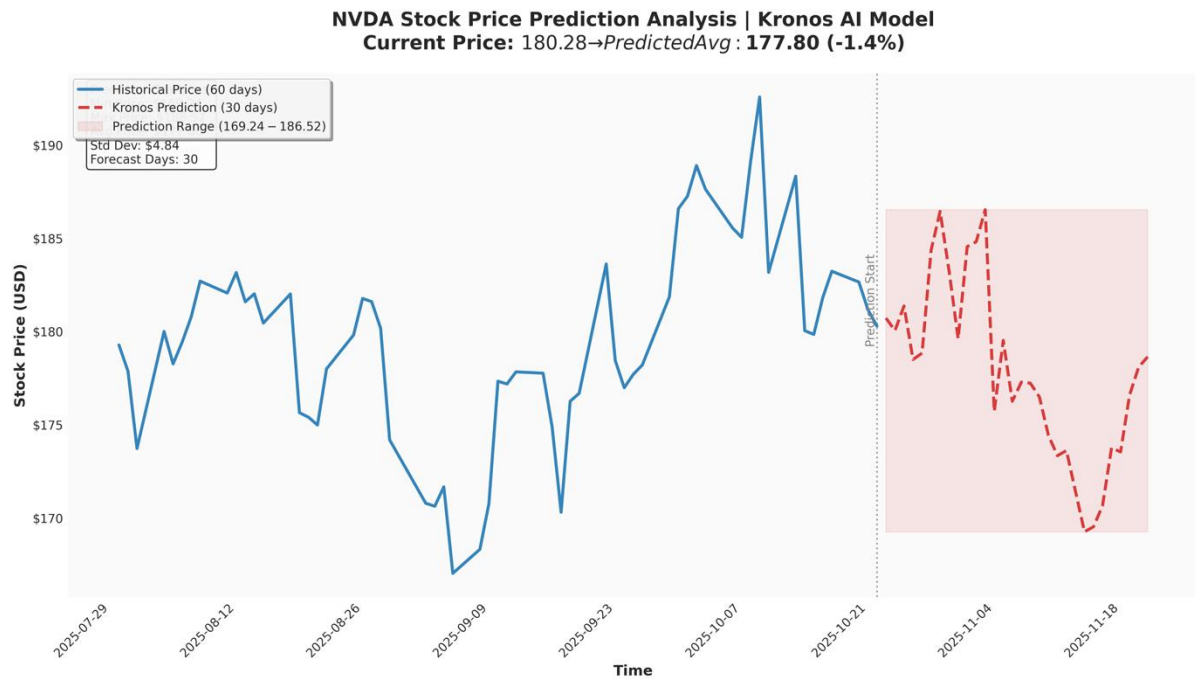


Figure 11 Kronos prediction result

(4) Risk Management Phase – Evaluation and Adjustment

The Risk Management module (risk_management/orchestrator.py) analyzes each Decision Card under three predefined profiles—Optimistic, Neutral, and Pessimistic.

It adjusts exposure, stop-loss, and target levels accordingly, and the Risk **Aggregator** synthesizes a unified risk report, ensuring decisions remain consistent with user tolerance levels.

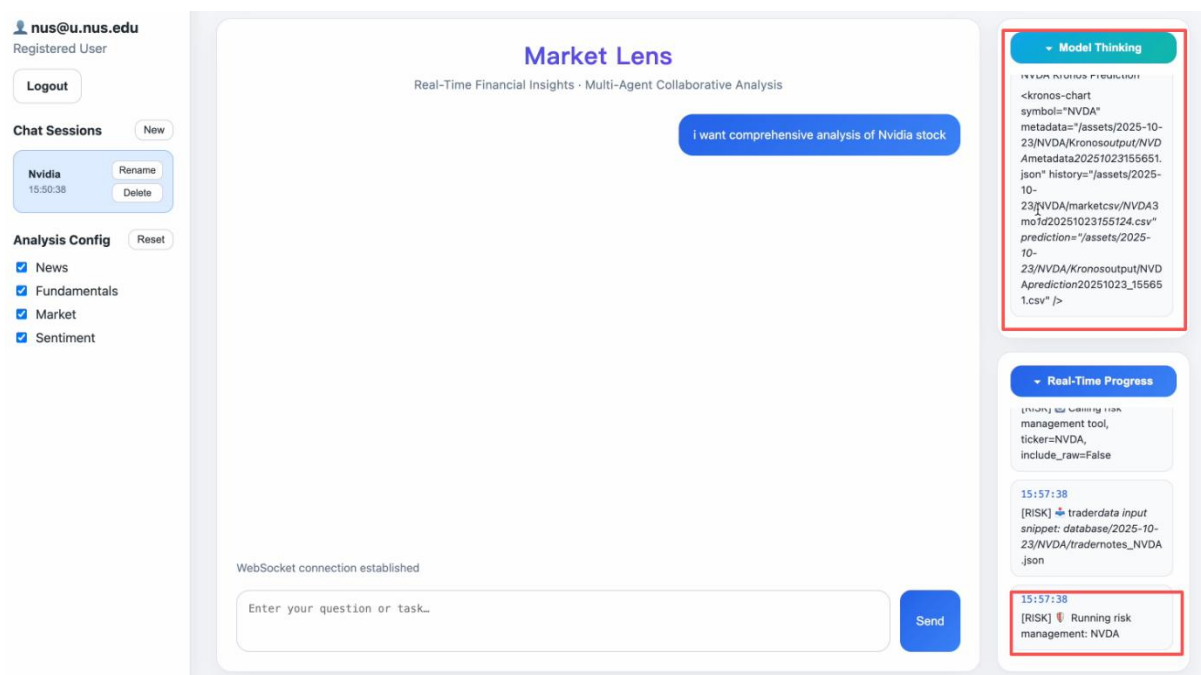


Figure 12 Risk Management

```

@dataclass
class RiskManagementOrchestrator:
    """Coordinate evaluators and produce the final risk report."""

    use_llm: bool = True
    max_workers: int = 3
    aggregator: RiskAggregator = field(default_factory=RiskAggregator)

    def _build_evaluators(self) -> List:
        return [
            OptimisticRiskEvaluator(use_llm=self.use_llm),
            NeutralRiskEvaluator(use_llm=self.use_llm),
            PessimisticRiskEvaluator(use_llm=self.use_llm),
        ]

    def _run_evaluators(self, card: TraderDecisionCard) -> List[RiskPerspective]:
        evaluators = self._build_evaluators()
        perspectives: List[RiskPerspective] = []

        if self.max_workers <= 1:
            for evaluator in evaluators:
                perspectives.append(evaluator.generate(card))
            return perspectives

        with ThreadPoolExecutor(max_workers=min(self.max_workers, len(evaluators))) as executor:
            future_map = {executor.submit(e.generate, card): e for e in evaluators}
            for future in as_completed(future_map):
                perspective = future.result()
                perspectives.append(perspective)

        # Preserve deterministic order: optimistic, neutral, pessimistic
        perspectives.sort(key=lambda p: {"optimistic": 0, "neutral": 1, "pessimistic": 2}[p.view_name])
        return perspectives

```

Figure 13 Orchestrator

(5) Result Delivery and Visualization

Finally, the Manager aggregates all outputs and returns the structured results to the interface, which displays the reasoning chain, charts, and Decision Cards interactively.

The interface also provides session persistence via a chat history manager, allowing users to revisit past analyses.

NVDA Risk module temporarily unavailable, proceed with caution. Trading recommendation summary: ### Trading Decision Card: NVIDIA (NVDA)

1. Research Synthesis & Strategy

The research file on NVIDIA (NVDA) has been processed. The analysis indicates a **SELL** recommendation with a confidence score of 0.75. This is primarily driven by significant geopolitical risks, namely the company's complete exit from the Chinese market, which creates a substantial long-term headwind for revenue. This fundamental concern is amplified by strongly bearish investor sentiment and adverse technical indicators.

Given the user's request for a "comprehensive analysis" and the research noting a "very_high" level of uncertainty, a Kronos forecast was executed to provide a forward-looking price outlook.

2. Kronos Prediction Analysis

A 30-day Kronos forecast was generated for NVDA. The key outputs are:

- **Mean Price:** 177.80
- **Max Price:** 186.52
- **Min Price:** 169.24

The forecast indicates a wide potential price range, which reflects the high uncertainty identified in the research. The mean predicted price is slightly below the current price of 180.28, which aligns with the overall bearish research stance.

3. Execution Plan

The following decision card has been generated, integrating the research conclusions and the Kronos forecast. The plan is to execute a SELL order, with risk controls set based on the current price and market conditions.

Decision Card: NVDA ``json

```
{
  "symbol": "NVDA",
  "decision": "SELL",
  "current_price": 180.28,
  "confidence_score": 0.75,
  "reasoning": "The recommendation to SELL is based on the research finding that positive factors such as market leadership and Dow Jones inclusion are outweighed by significant risks. Key concerns include NVIDIA's complete exit from the Chinese market, which poses a long-term threat to revenue, strongly bearish investor sentiment, and negative technical indicators (price below 20-day MA, RSI below 40). Furthermore, the absence of EPS data complicates a full fundamental valuation. The Kronos prediction was invoked due to the 'very_high' uncertainty, and its forecast aligns with a cautious to bearish outlook.",
  "has_prediction": false,
  "position_size": {
    "percentage": 0.075,

```

4. Risk Management Overlay

A secondary risk analysis was performed on the trading plan. The review suggests a more conservative approach due to the identified risks. The adjusted parameters are:

- **Adjusted Confidence:** 0.58
- **Adjusted Position Size:** 4.3%

This suggests reducing the position size from the initial 7.5% to 4.3% to better manage potential downside risk.

Final Recommendation

Based on the comprehensive analysis, the final recommendation is to **SELL NVDA** with a reduced position size of **4.3%**. The stop-loss is set at **\$189.29** and the take-profit target is **\$153.24**.

NVDA Kronos Prediction

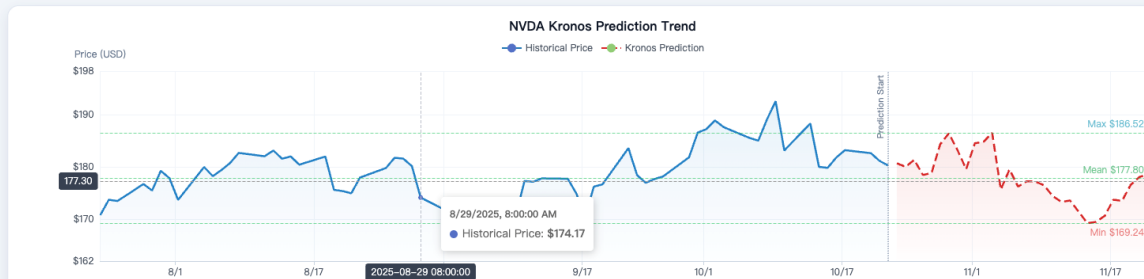


Figure 14 Final Answer

After the report is generated, users can continue to inquire about any stock-related information in the same session.

The Manager Agent utilizes its contextual memory and previously stored analysis files to provide informed, context-aware responses.

When existing information is insufficient, the Manager automatically invokes relevant tools or sub-agents to retrieve additional real-time data or perform supplementary analysis, ensuring that follow-up queries remain accurate, dynamic, and up to date.

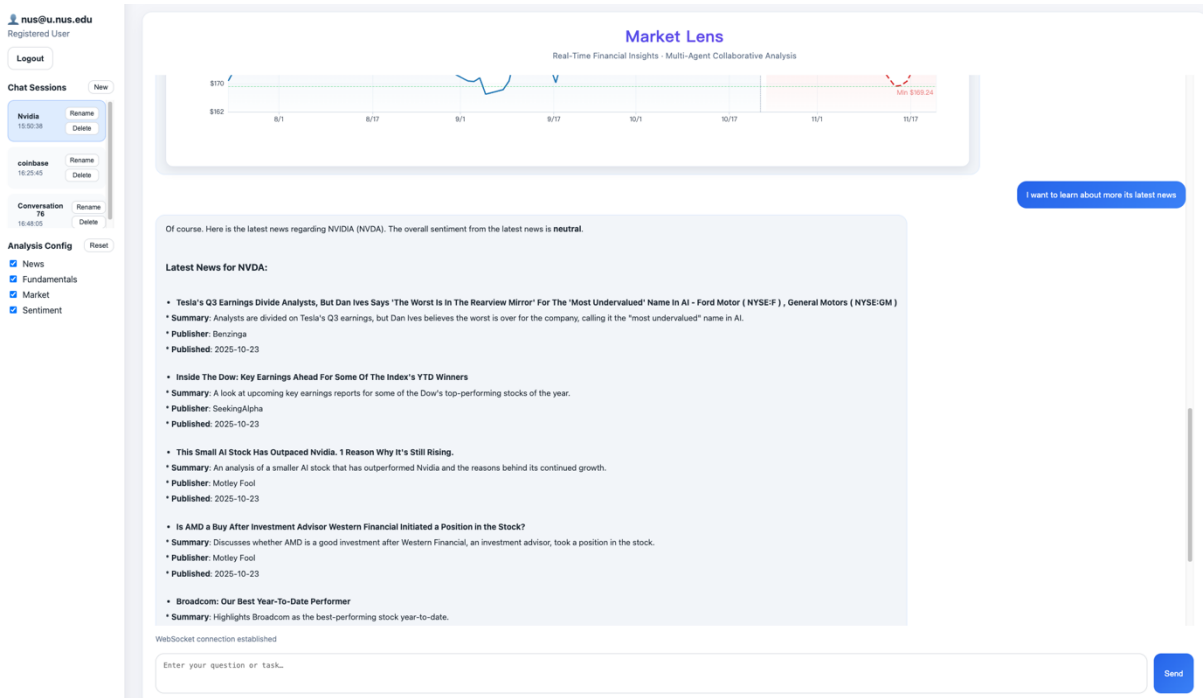


Figure 15

7.2 Implementation Progress

All major modules have reached functional completion and are operating cohesively. Internal testing confirms that each agent executes correctly when called sequentially by the Manager.

Table 1 Integration milestones

Module	Core Functionality	Status	Remarks
Analyst	Unified multi-source analysis (news, fundamentals, market, sentiment)	✓	Includes data caching and CSV output
	Multi-round reasoning and moderation	✓	Structured JSON reasoning outputs
Trader	Decision Card generation with Kronos model integration	✓	Produces visual predictions and trade proposals
	Three-profile evaluation and synthesis	✓	Validates trade safety and exposure limits
React + FastAPI Frontend	Chat-based control and visualization	✓	Supports live result rendering and session management
Backtesting	Strategy evaluation and performance metrics	🕒 In progress	Planned integration with Kronos outputs

7.3 Technical Stack and Tools Used

The Market Lens system integrates a modern, asynchronous AI reasoning architecture with robust frontend and backend technologies. This hybrid stack supports both academic transparency and market-grade scalability.

Table 2 System Architecture Layers and Technology Stack

Layer	Technology/ Framework	Functionality and Role
Analyst Frontend	React (TypeScript)	Builds a dynamic SPA for user queries, reasoning visualization, and chart rendering.
	WebSocket (Socket.IO / native API)	Enables low-latency, bi-directional data streaming for reasoning progress and results.
		Orchestrates requests, handles REST + WebSocket communication, manages task scheduling, and validates responses using Pydantic.
Backend Gateway	FastAPI	
Reasoning Framework	Python 3.11 + LangChain	Manages multi-agent reasoning (Manager, Analyst, Researcher, Trader, Risk Manager).
Models and AI Agents	Gmini-based LLMs, Kronos forecasting model	Supports reasoning, debate, summarization, and short-term quantitative prediction.
Data Sources	Yahoo Finance, Finnhub, FMP, Alpha Vantage, Playwright	Retrieves market, news, and sentiment data through

		APIs and automated web scraping.
Visualization	Plotly / Recharts, Tailwind CSS, WebSocket streaming	Renders reasoning chains, risk heatmaps, and dynamic chart visualizations.

This combination ensures robust reasoning performance, real-time streaming, and explainable AI integration across the Market Lens platform.

7.4 Technical Challenges

7.4.1 Foundation Model Selection and Reasoning Efficiency

At the initial stage of development, we relied on the GPT series (via paid API) as the backbone reasoning model. While the model demonstrated strong reasoning accuracy and interpretability, the token consumption rate was prohibitively high, rendering large-scale experimentation financially unsustainable.

To reduce operational costs and enable on-premise deployment, we transitioned to QWQ-32B, a locally hosted large language model. However, this model lacked function-calling capabilities, which are fundamental to our tool-oriented reasoning framework—particularly for invoking analytical utilities such as data retrieval, financial metric computation, and risk evaluation.

Subsequently, we adopted Qwen/Qwen3-235B-A22B, which offered enhanced reasoning capabilities and partial support for tool use. Despite this improvement, we encountered frequent tool-invocation errors and recursive call deadlocks during multi-agent coordination. After in-depth diagnostics, we identified the root cause as the model’s limited context length in “thinking mode,” which led to truncated reasoning memory and infinite recursive loops.

To stabilize the pipeline, we temporarily reverted to a GPT-4-based reasoning setup, which successfully restored functional integrity. However, this configuration introduced latency in sub-agent communication, primarily because each sub-agent returned entire data payloads rather than summarized results—placing excessive demand on the context window and slowing inter-agent reasoning.

To overcome this issue, we redesigned the data-exchange mechanism:

- Each sub-agent now saves intermediate outputs as local JSON files.
- Instead of transmitting raw data, only the file path is returned to the Manager Agent.

- The Manager was upgraded with a read-file capability, allowing it to autonomously retrieve relevant information on demand when additional context is required.

This redesign significantly reduced memory overhead, improved I/O efficiency, and accelerated agent-to-agent communication, while enabling the Manager to operate with greater autonomy and contextual awareness.

Looking ahead, we plan to migrate to the Gemini family of models, which provide multi-million-token context windows and transparent thinking-trace visualization. These features are crucial for achieving scalable, explainable, and continuous multi-agent reasoning—advancing both the system’s academic rigor and its practical performance.

7.4.2 Frontend Technology Stack Evolution

The initial prototype was built using a FastAPI + WebSocket + React framework for rapid development of the chat-based user interface. While this setup enabled efficient communication between the frontend and backend, it initially presented several limitations in dynamic UI updates and real-time data synchronization, requiring further optimization in event handling and rendering performance.

Specifically, UI elements inside the `st.spinner()` context could not be updated in real time; the interface refreshed only after all backend computations had completed.

This behavior prevented the display of real-time reasoning logs, progress indicators, and dynamic charts, which are core to Market Lens’s explainable design philosophy.

To overcome this constraint, the frontend architecture was completely refactored into a React + FastAPI + WebSocket framework.

This migration provided several benefits:

- **Real-time bi-directional communication**, allowing the system to stream intermediate “thinking” content from agents.
- **Component-based chart rendering**, enabling live visualization of technical indicators, reasoning chains, and confidence metrics.
- **Scalable UI updates**, supporting progress monitoring and multi-agent output visualization concurrently.

The resulting architecture not only improved responsiveness and interactivity, but also laid the foundation for future multi-session support and front-end integration with mobile clients.

7.4.3 Sentiment Data Collection and Cost Optimization

For sentiment analysis within the Analyst module, we initially planned to access real-time Twitter/X API feeds to capture market mood and financial discussions.

However, the API access costs for streaming endpoints were significantly higher than those for standard financial-data APIs, which exceeded our project's budget constraints.

To maintain data freshness while reducing expenses, we implemented an automated web-scraping pipeline using Playwright.

In this approach:

- We launch a headless browser on the backend.
- We load multiple authenticated sessions (via stored cookies) to bypass login limitations.
- The browser then automatically navigates to Twitter search pages, queries relevant ticker symbols and financial keywords, and captures textual content for NLP-based sentiment extraction.

This method achieved comparable coverage and timeliness to API-based solutions while reducing operational costs by over 90%.

We further process the scraped data through intelligent deduplication and sentiment-scoring pipelines, ensuring the quality, reliability, and diversity of the social signals used in reasoning.

8 Results and Progress

8.1 Preliminary Results and Module Outcomes

This section presents the observed outcomes from the integrated Market Lens system. Each agent has been functionally tested within the full pipeline, from multi-source analysis to risk evaluation. The preliminary results demonstrate that the system successfully transforms raw financial data into structured, explainable, and risk-adjusted investment insights.

8.1.1 Analyst Layer – Evidence-Based Market Understanding

During testing, the Analyst module produced consistent and structured outputs by combining financial news, fundamentals, market data, and sentiment.

Each analysis cycle resulted in a unified JSON summary containing sentiment scores, technical indicators, and financial metrics.

Representative outcomes include:

- **News Summarization:** Gemini-based extraction generated concise summaries of market-moving headlines, with sentiment orientation automatically classified.
- **Fundamental Metrics:** Retrieved valuation ratios and profitability indicators (P/E, EPS growth, debt ratio), ensuring fallback data continuity.
- **Market Indicators:** Technical analysis (RSI, MACD, Bollinger Bands) captured short-term overbought/oversold trends.
- **Sentiment Index:** Computed from weighted Twitter data, providing an early warning of sentiment shifts preceding price changes.

Table 3 Technical Indicators Summary (AAPL) — As of 2025-10-23

Indicator	Current Value	Interpretation
RSI	51.6	Neutral market momentum
20-Day Moving Average (MA20)	\$254.52	Current price slightly above short-term average
Bollinger Bands	[\$244.78, \$264.27]	Price near upper band — mild upward pressure
ATR (%)	2.13%	Moderate short-term volatility
20-Day Trend	+0.6%	Slight upward trend

Table 4 Social Media Sentiment Summary (AAPL) — As of 2025-10-23

Category	Details
Overall Sentiment	Strong Bullish (Highly Optimistic)
Sentiment Score	0.4993
Total Tweets	80
Bullish (%)	16.3%
Bearish (%)	3.7%
Neutral (%)	80%

Table 5 Representative Opinions

Sentiment	Example Tweet
Bullish	“Apple hits a new all-time high! Analysts call it the start of a long-awaited upgrade cycle.”
Neutral	“Big Tech’s Q3 earnings season could reshape 2025.”
Bearish	“iPhone Air has fallen short of expectations.”

8.1.2 Researcher Layer – Multi-Round Reasoning and Consensus Building

The Researcher layer effectively executed the multi-round debate mechanism between Bullish and Bearish reasoning tools.

Each round produced a structured argument of records containing key points, counterarguments, and referenced evidence.

The moderation phase consolidated them into a final conclusion with confidence weighting.

Table 6 Researcher Debate Summary (AAPL) — As of 2025-10-23

Round	Description	Participants	Key Focus	Outcome
Round 1	Initial debate with differing views — some analysts were bullish due to strong iPhone 17 sales and large buyback program, while others remained cautious over valuation.	Long-side Analyst Neutral Analyst Short-side Analyst	Fundamentals vs. Valuation	Mixed opinions — no consensus

Round	Description	Participants	Key Focus	Outcome
Round 2	Discussion shifted to missing fundamental data (EPS, debt), with Short-side emphasizing reduced transparency and valuation risk.	Same participants	Data transparency & risk exposure	Leaning bearish sentiment
Round 3	Final synthesis of all perspectives; aggregator combined all arguments and computed weighted confidence.	Aggregator + all analysts	Consensus building	Final Verdict: SELL (Medium-Term), Confidence: 0.6

8.1.3 Trader Layer – Strategy Formation with Kronos Forecasting

The Trader module, powered by the Kronos prediction engine, successfully generated executable Decision Cards that integrate both qualitative reasoning and quantitative forecasts. Each Decision Card includes operation type, entry/exit range, stop-loss, target, and confidence. Kronos model tests historical price data produced coherent short-term forecasts, with predicted K-line curves closely matching actual price trends.

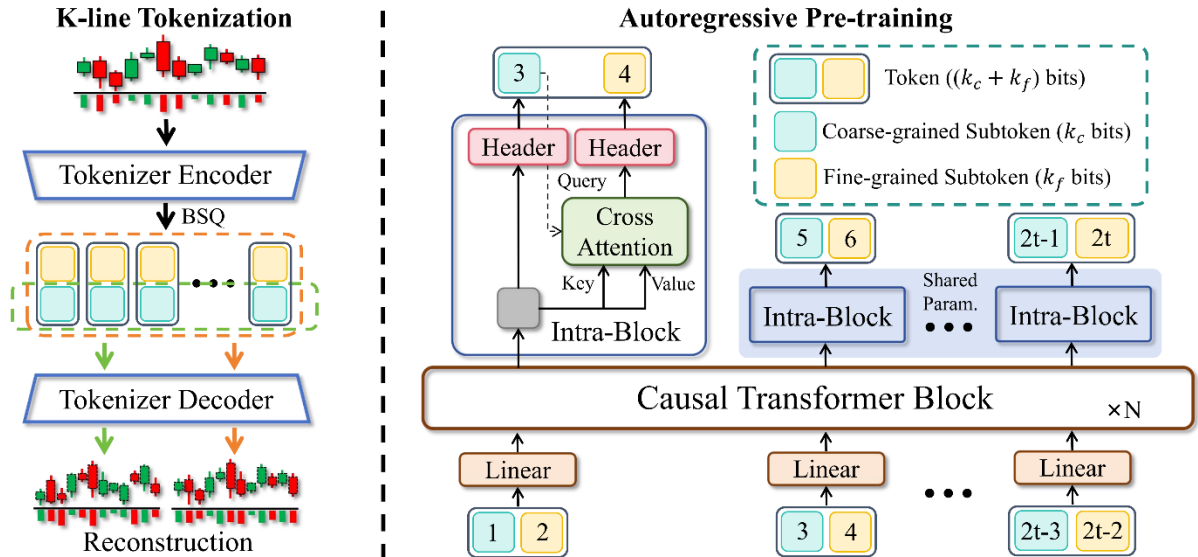


Figure 16 the Structure of Kronos

Table 7 Trading Decision Card (AAPL) — As of 2025-10-23

Category	Details
Overall Recommendation	SELL
Confidence Level	0.6 (Medium Confidence)
Data Sources	Researcher Module + Kronos Predictor
Rationale	Missing key financials (EPS, debt); weak innovation (iPhone Air underperforming); valuation near 52-week high; high concentration risk in mega-caps (AAPL, NVDA, MSFT).
Model Validation (Kronos)	Predicted 30-day mean price: \$247.85 — lower than current \$258.45. Predicted range: \$226.04 – \$275.24 → supports bearish bias.

Table 8 Execution Plan (AAPL) — As of 2025-10-23

Parameter	Value	Explanation
Decision	SELL	System aligns with research and model output
Symbol	AAPL	Apple Inc.
Confidence	0.6	Moderate certainty
Position Size	6.0 % of portfolio	Controlled exposure to downside risk
Execution Range	\$253.28 – \$263.62	± 2 % from current price
Stop-Loss	\$271.37	Protects against upward reversal
Take-Profit	\$219.68	Locks in expected downside gain

Table 9 Interpretation (AAPL) — As of 2025-10-23

Aspect	Insight
Market View	Short-term downside expected — likely mild correction from overvaluation zone
Strategy Type	Medium-term defensive trade
System Behavior	Integrates research debate outcome and Kronos deep prediction to form a unified decision card with risk controls

8.1.4 Risk Management Layer – Multi-Perspective Evaluation

The Risk Management module validated the safety and consistency of each trading proposal through three concurrent evaluation profiles—Optimistic, Neutral, and Pessimistic.

The evaluators automatically scaled position sizing and adjusted stop-loss and take-profit thresholds according to risk level.

Table 10 Risk Perspectives (AAPL) — As of 2025-10-23

Perspective	Summary	Position Size	Stop-Loss	Take-Profit	Confidence
Optimistic	Treats the SELL signal as a tactical pullback opportunity rather than a long-term bearish view. Main risk: missing a rebound.	0.40	\$265.00	\$254.00	0.70
Neutral	Notes missing stop-loss/take-profit in the trader card; system derives symmetric risk bands automatically.	0.50	\$264.00	\$252.90	0.60
Pessimistic	Focuses on capital protection and rapid de-risking (de-leveraging). Assumes rising volatility.	0.35	\$266.20	\$211.93	0.59

Table 11 Trader vs. Risk Management (AAPL) — As of 2025-10-23

Aspect	Trader	Risk Management	Change / Improvement
Decision	SELL	SELL (confirmed)	Consistent, validated
Position Size	0.6 (6%)	0.417 (4.17%)	↓ ~30% exposure
Stop-Loss	\$271.37	\$264.00	Tighter stop
Take-Profit	\$219.68	\$211.93	More conservative
Confidence	0.60	0.63	Slightly higher
Risk Budget	—	High	Defined explicitly
Extras	—	Added hedging + alerts	Improved control & safety

8.1.5 Frontend Visualization and Interaction

The new React + FastAPI interface provides a real-time, bi-directional communication channel between user and system through WebSocket streaming. Users can now observe live execution logs, progressive responses, and detailed agent interactions as they occur.

- Natural language commands triggered multi-agent execution.
- Generated charts, decision cards, and reasoning summaries were visualized in real time.

- A dedicated log panel streams execution traces from all agents in real time, showing when each sub-agent starts, completes, or encounters errors.
- Users can monitor the reasoning process dynamically, improving transparency and engagement.

Please refer to **Section 7.1 Workflow Overview** and the **System Video** for detailed front-end visuals.

8.2 Deviation from Initial Plan

While the Market Lens project closely followed the original roadmap, several design adjustments and development refinements were introduced during implementation.

These deviations were driven by opportunities for optimization and technical enhancement rather than delays, which resulted in a more efficient, explainable, and academically robust system.

Table 12 Summary of Deviations

Aspect	Initial Plan	Actual Implementation	Type of Deviation
Trader Module Logic	Planned use of a rule-based strategy	Replaced with the Kronos deep learning model,	Positive enhancement
	generator integrating research outcomes	enabling predictive time-series forecasting	
Analyst Layer Design	Initially designed as four independent sub-agents	Consolidated into a single LangChain agent with modular tools for data retrieval and analysis	Architectural refinement
Development Schedule	Modules planned for sequential development (Analyst → Researcher → Trader → Risk Manager)	All core modules were developed in parallel to accelerate integration and maintain consistency	Parallel development
Frontend Framework	Planned implementation with FastAPI, WebSocket, and React for	Upgraded to React + FastAPI + WebSocket architecture supporting real-time streaming and	Major architectural enhancement

	interaction and visualization.	multi-session management	
Backtesting Component	Scheduled for full integration by final phase	Still in progress, partially linked with Kronos predictions and performance metrics	Minor delay

8.2.1 Key Reasons for Deviations

(1) Technical Optimization

The adoption of the Kronos forecasting engine replaced the initial rule-based trader logic, enhancing accuracy and aligning the project more closely with academic goals in decision automation and machine reasoning.

(2) Efficiency through Parallel Development

To improve workflow efficiency, all core modules—Analyst, Researcher, Trader, and Risk Management—were developed concurrently.

This approach enabled earlier end-to-end integration testing and faster debugging without compromising output consistency.

(3) Architectural Simplification

The consolidation of four Analyst sub-agents into a single modular LangChain agent reduced system overhead and improved performance stability.

(4) User-Centric Adjustment

The switch to React and WebSocket architecture was driven by the need for real-time feedback and improved scalability, enabling live streaming of reasoning processes and execution logs.

8.2.2 Impact on Academic and Market Relevance

Academic Impact:

- The deviations strengthened alignment with MR–RS–CGS frameworks.
- MR: The multi-round debate and Kronos predictive reasoning enhanced cognitive depth.
- RS: Streamlined orchestration and automated risk synthesis reinforced structured reasoning.
- CGS: Parallelized modules and live feedback in the frontend improved human–AI collaboration transparency.
- As a result, the system offers a more concrete demonstration of reasoning systems in applied finance.

Market Impact:

- Earlier deployment of the visualization layer improved user experience and interpretability, both essential for market adoption.
- Integration of Kronos forecasting introduced realistic quantitative validation, improving the model's relevance to trading applications.
- The risk management module's parallelization ensures scalability and adaptability for future real-time trading use cases.

8.3 Relation of Results to Academic and Market Expectations

The current progress of the Market Lens system demonstrates that multi-agent reasoning can effectively transform scattered financial information into structured, explainable, and actionable insights.

Beyond achieving technical functionality, the project delivers meaningful contributions to both academic research and market applications.

8.3.1 Relation to Academic Expectations

The Market Lens framework operationalizes theoretical concepts from Machine Reasoning (MR), Reasoning Systems (RS), and Cognitive Systems (CGS) through its layered agent design:

- **Machine Reasoning (MR):**

The Researcher layer embodies argument-based reasoning through multi-round debates between bullish and bearish perspectives. This demonstrates how LLM-driven agents can reason over conflicting evidence and synthesize transparent, justifiable conclusions.

- **Reasoning Systems (RS):**

The Trader and Risk Management layers showcase automated decision generation under defined constraints. The integration of the Kronos model bridges symbolic reasoning and quantitative prediction, demonstrating practical decision automation in finance.

- **Cognitive Systems (CGS):**

The Manager Agent and interactive React + FastAPI interface enable seamless human–AI collaboration. Users can trace how each decision is formed, verify reasoning steps, and adjust system parameters—transforming abstract cognitive principles into practical, interpretable interactions.

8.3.2 Relation to Market Expectations

From a market standpoint, Market Lens demonstrates the potential of explainable AI in financial decision support:

- **Clarity and Accessibility:**

The system distills complex market signals—across news, fundamentals, and sentiment—into interpretable outputs that reduce cognitive load for non-expert investors.

- **Transparency and Trust:**

The adoption of a real-time React + WebSocket frontend significantly enhances market applicability by offering continuous visibility into reasoning and trading decisions—an essential feature for user trust and operational transparency in financial AI systems.

- **Risk-Aware Automation:**

Multi-perspective evaluation (Optimistic, Neutral, Pessimistic) provides a balanced framework for automated strategy review, aligning with real-world portfolio management principles.

- **Practical Adaptability:**

The modular, API-based design allows the framework to evolve toward integration with live data feeds or external analysis tools, enhancing future scalability.

9 Challenges and Roadblocks

While Market Lens has reached a functionally complete state, several technical and conceptual challenges remain. These issues influence both the academic validity of the reasoning framework and its potential market scalability. This section identifies these obstacles and outlines strategies for resolving them within the remaining project period.

9.1 Technical and Implementation Challenges

Real-Time Data Integration

- Problem: Financial APIs differ in latency and data formats, complicating synchronization across the multi-agent workflow.
- Academic Implication: Inconsistent input streams may affect reasoning reproducibility and evaluation of reliability.
- Strategy: Employ asynchronous task scheduling with local caching and define fixed analysis windows to maintain temporal consistency.

Multi-Agent Coordination Complexity

- Problem: Coordination among multiple LangChain agents occasionally leads to delayed responses or dependency deadlocks.
- Academic Implication: Highlights the need to test reasoning orchestration robustness within complex agent ecosystems.
- Strategy: Introduce a centralized task queue and retry mechanism within the Manager Agent, reinforce data validation in JSON schema.

Explainability and Evaluation Metrics

- Problem: Reasoning outputs are detailed but sometimes overly verbose, making it hard for users to extract key insights.
- Academic Implication: Limits practical interpretability and constrains formal assessment of reasoning quality.
- Strategy: Design a concise “Reasoning Summary” interface and implement a coherence score to measure logical consistency.

9.2 Data and Market Application Challenges

Data Reliability and Coverage

- Problem: Public APIs (Yahoo, Finnhub, Reddit) occasionally experience outages or inconsistent updates.

- Market Implication: Affects real-time accuracy and user confidence in generated trading signals.
- Strategy: Use fallback APIs, implement redundancy in the Analyst layer, and store periodic snapshots to prevent data loss.

User Scalability and Deployment Performance

- Problem: Current implementation performs well in single-user scenarios but may face latency under concurrent usage.
- Market Implication: Limits scalability and commercial viability of the prototype.
- Strategy: Introduce lightweight backend services (FastAPI with WebSocket) and enable React-side state caching for improved responsiveness and overall system efficiency.

Risk Interpretation for Non-Expert Users

- Problem: Multi-profile risk reports (Optimistic, Neutral, Pessimistic) are complex for general investors to interpret.
- Market Implication: Reduces system accessibility for retail users.
- Strategy: Add simplified risk visualizations (e.g., heatmaps, risk-level labels) and plain-language summaries.

9.3 Academic and Conceptual Roadblocks

Evaluation of Reasoning Validity

- Problem: There is no standardized metric to evaluate the correctness of reasoning in multi-agent debates.
- Academic Implication: Makes it difficult to objectively validate cognitive reasoning quality.
- Strategy: Conduct expert-based qualitative reviews comparing debate outputs with factual financial outcomes.

Balancing Autonomy and Control

- Problem: Managing trade-offs between user oversight and agent autonomy remains challenging.
- Academic Implication: Directly relates to Cognitive Systems research on human–AI collaboration and controllability.
- Strategy: Retain dual operation modes (automatic/manual) and expose reasoning logs for transparency.

9.4 Mitigation Strategies and Timeline Alignment

To ensure progress toward the final submission, the team has planned the following mitigation actions:

- Stabilize the real-time data pipeline and caching mechanisms.
- Streamline multi-agent task orchestration and error recovery processes.
- Enhance visualization clarity for non-technical users through improved front-end summaries.
- Maintain academic validation by collecting qualitative feedback on reasoning outputs from domain experts.

These actions will be executed in parallel during the remaining development phase to strengthen both the academic credibility and market readiness of the Market Lens system.

10 Future Work

The current version of Market Lens has successfully established a fully functional prototype that validates the core concept of a multi-agent reasoning system for investment intelligence. Building upon this foundation, our future work is planned across short-term refinements and long-term strategic expansions, aimed at enhancing both the system's robustness and its practical impact.

10.1 Short-Term Goals and Immediate Next Steps

Our immediate focus is on polishing the user experience and ensuring the system's technical stability for broader demonstration and testing purposes.

- **User Interface (UI) Refinement and Enhancement:** We plan to conduct a comprehensive redesign of the React + FastAPI-based web interface to further enhance its aesthetic appeal, interactivity, and usability. This involves developing a modern, responsive layout, refining the information hierarchy within Decision Cards, and improving the visualization of reasoning chains and risk reports to enable faster comprehension and a more intuitive user experience for end-users.
- **Scalability and Concurrent User Support:** The current system architecture is optimized for single-user or low-concurrency scenarios. A key short-term goal is to re-architect the backend service to efficiently handle higher volumes of concurrent requests. This will involve implementing a more robust task queue (e.g., using Celery with Redis/RabbitMQ), introducing session caching mechanisms, and potentially decoupling the frontend and backend using lightweight frameworks like FastAPI. This upgrade is crucial for stress testing and preparing the system for any potential pilot programs or larger-scale academic demonstrations.

10.2 Long-Term Vision and Strategic Roadmap

Looking beyond the immediate academic scope of the project, the long-term vision for Market Lens is to evolve into a robust, scalable, and user-centric platform that bridges the gap between academic research and real-world application.

- **Expansion and Fine-tuning of Prediction Models:** We plan to conduct a comprehensive redesign of the React + FastAPI-based web interface to enhance its visual appeal, interactivity, and overall usability. This includes implementing a more modern and intuitive layout, optimizing the information hierarchy within Decision Cards, and

improving the visualization of reasoning chains and risk reports to enable faster comprehension and more engaging user experiences.

- **Quantitative Strategy Integration and Alpha Combination:** A significant evolution will be the introduction of professional quantitative forecasting capabilities. This involves moving beyond single-signal recommendations to constructing multi-factor models that combine various alpha sources (e.g., momentum, value, quality, sentiment) generated by the AI agents. The system will aim to automatically build, weight, and rebalance integrated investment strategies, leveraging the multi-agent framework to dynamically assess and synthesize diverse predictive signals into a more robust and adaptive portfolio of insights.
- **Backtesting Framework and Automated Execution:** To ensure the practical efficacy and continuous improvement of generated strategies, we plan to develop a sophisticated backtesting engine. This engine will allow for rigorous historical simulation of Decision Cards and quantitative strategies against comprehensive market data, providing key performance metrics (e.g., Sharpe ratio, max drawdown, alpha). The insights from backtesting will create a feedback loop for iteratively refining the AI agents' reasoning and strategy generation logic. Ultimately, this capability paves the way for integration with major financial exchanges via APIs, enabling paper trading and, subsequently, automated execution of vetted strategies in live markets, thereby closing the loop from insight to action.
- **Mobile Application Development:** To maximize accessibility and align with modern user habits, a primary long-term goal is the development of a native mobile application for iOS and Android platforms. This app will deliver the core functionality of the web platform—allowing users to receive AI-generated investment insights, review Decision Cards, and monitor their simulated or real portfolios—in a convenient, on-the-go format. This step is critical for transitioning Market Lens from a technical prototype to a tangible product that can seamlessly integrate into an investor's daily routine.

By pursuing this roadmap, Market Lens will not only solidify its academic contributions to intelligent reasoning systems but also advance its potential to become a practical tool that empowers investors with transparent, evidence-based intelligence.

11 Conclusion

11.1 Summary of Project Highlights

Market Lens has successfully conceptualized, designed, and implemented a functional prototype of a multi-agent financial intelligence system. The project's primary achievement lies in its novel integration of multiple large language model (LLM) agents, each emulating a specialized role within an investment research team, into a cohesive and automated reasoning pipeline. Key technical highlights include the development of a robust data ingestion layer aggregating multi-source financial data; the implementation of a multi-round adversarial debate mechanism within the Researcher agent, which synthesizes bullish and bearish perspectives into evidence-based conclusions; the generation of structured and interpretable Decision Cards by the Trader agent, enhanced by the integration of the Kronos forecasting model; and a dynamic risk management framework that evaluates strategies across multiple risk profiles. The entire system is accessible through an interactive web interface, demonstrating a closed-loop workflow from user query to actionable, risk-adjusted insight.

11.2 Key Academic Insights and Potential Market Value

From an academic perspective, Market Lens serves as a concrete implementation and validation of core concepts taught in Machine Reasoning (MR), Reasoning Systems (RS), and Cognitive Systems (CGS). It demonstrates the practical application of multi-agent reasoning architectures for complex, real-world decision-making under uncertainty. The project contributes insights into how structured debate and moderation can enhance the robustness and explainability of LLM outputs, while the integration of symbolic reasoning (agent debates) with quantitative forecasting (Kronos model) explores a hybrid approach to intelligent system design.

In terms of market value, this project directly addresses a significant gap in the current landscape of retail investment tools. By transforming fragmented and overwhelming market data into clear, evidence-backed insights, Market Lens offers a compelling value proposition: to democratize access to institutional-grade analysis. Its emphasis on transparency, through auditable reasoning chains and risk annotations, tackles the critical issues of trust and explainability that often hinder the adoption of AI-driven financial tools. The system demonstrates the potential to reduce cognitive load on investors, mitigate emotional decision-making, and promote a more disciplined, research-oriented investment approach.

11.3 Reflection on Learning Outcomes and Practical Impact

The development of Market Lens provided invaluable practical experience in architecting and integrating a complex, intelligent system. It deepened our understanding of the entire lifecycle of an AI project, from data acquisition and agent-based modeling to front-end visualization and user-centric design. A key learning outcome was navigating the challenges of ensuring reliability and coherence in a multi-component pipeline, where the output of one agent serves as the input for another. Furthermore, the project underscored the importance of balancing ambitious technical goals with pragmatic constraints, leading to informed design decisions that optimized for functionality and academic rigor.

The practical impact of Market Lens extends beyond its technical specifications. It stands as a proof-of-concept for how AI can be leveraged not to replace human judgment, but to augment it by providing structured, transparent, and reasoned analysis. This project highlights a promising direction for the future of decision-support systems, not only in finance but in any domain characterized by information overload and complex reasoning requirements. Market Lens successfully bridges theoretical AI concepts with a tangible application, showcasing the potential to make sophisticated analytical power more accessible and understandable.

- Summary of project highlights
- Key academic insights and/or potential market value
- Reflection on learning outcomes and practical impact

12. Supplementary Materials

12.1 Appendices

12.1.1 Video Prompt

Table 13 Video Prompt

Prompt	Description	Camera / Lighting / Mood Details	Video Reference
1	In a blue room, the CNA business news is reporting on the Singapore stock market trend of the day. One screen shows a man presenting the news, while another displays the K-line chart. As the camera pans right, two men in suits are analyzing the charts.	—	0:00 – 0:05
2	The camera slowly zooms out, revealing a young man in a grey hoodie sitting before multiple monitors showing trading charts, tickers, and numbers. The cool blue lighting comes mainly from the monitors.	—	0:09 – 0:16
3	As the zoom completes, red and green lights flicker across his face — showing stress and confusion. His eyes dart between screens.	Camera: Macro close-up → medium shot; slight handheld movement. Lighting: Red/green flickers, soft blue ambient light, shadow gradient. Mood: Tense, chaotic, immersive — conveying market pressure.	0:42 – 0:48

12.1.2 Code Prompt

Table 14 System Behavior Improvements

Prompt	Goal / Description	Key Enhancement
1. Logging System Enhancement	Make the system show real-time thinking progress instead of static logs.	Display “responded” text as live manager thoughts above the chat — updating dynamically as tasks run.
2. Real-time Sub-Agent Feedback	Add visible feedback while sub-agents are running.	Enable real-time streaming updates for each step, with smooth progress visualization on the frontend.
3. Kronos Visualization Integration	Automatically show Kronos prediction results visually.	Embed animated stock charts in chat for live display when predictions complete.

Table 15 Code Implementation References

Prompt	Related Modules / Files	Focus Area
1	agent_steam_gradio.py	Manager’s real-time progress display
2	orchestrator.py, debate.py, stream.py, useStream.ts	Backend streaming and frontend visualization flow
3	trader.py, index.html	Dynamic chart rendering and UI integration

12.1.3. Other GPT use

Table 16 Project References & Chat Records

Category	Description	Link
Report Review ①	Initial review and feedback discussion	View Chat
Report Review ②	Follow-up review and refinement session	View Chat
Report Review ③	Extended report analysis continuation	View Chat
Structure of Report	Discussion on overall report structure and organization	View Chat
Project Brainstorming	Brainstorming and discussing project details with GPT	View Chat

12.2 Mapped System Functionalities

This appendix maps the functionalities of the Market Lens system to the knowledge, techniques, and skills covered in the Intelligent Reasoning Systems modular courses: Machine Reasoning (MR), Reasoning Systems (RS), and Cognitive Systems (CGS). The mapping demonstrates how the system embodies core concepts from these courses, leveraging multi-agent reasoning, knowledge representation, and cognitive architectures to transform financial data into actionable insights.

12.2.1 Mapping Table

The following table summarizes the alignment between Market Lens functionalities and the MR, RS, and CGS course components. Each system module is described in terms of its key activities, followed by the corresponding course concepts it utilizes.

Table 17 Mapping Table

Market Lens Module	Key Functionalities	MR Course Alignment	RS Course Alignment	CGS Course Alignment
Analyst Layer	- Data aggregation from multiple sources (market data, news, fundamentals, sentiment)	Search Techniques (MR Day 1): Uninformed and informed search (e.g., Dijkstra's algorithm) for efficient data retrieval and filtering.	Rule-Based Reasoning (RS Day 2): Use of rules for data cleaning and normalization.	LLM Reasoning (CGS): Application of NLP models (e.g., Gemini) for text summarization and sentiment extraction, akin to Chain-of-Thought prompting.
	- Preprocessing and normalization			
	- Technical indicator computation (e.g., RSI, MACD)			
	- Sentiment analysis using NLP			
		Knowledge Representation (MR Day 2): Structured data representation using JSON and semantic networks for interoperability.	Probabilistic Reasoning (RS Day 2): Handling uncertainty in sentiment data via Bayesian methods.	

Researcher Layer	- Multi-round debates between bullish and bearish agents - Consensus building with confidence scoring - Evidence-based conclusion synthesis (BUY/SELL/HOLD)	Deductive and Abductive Reasoning (MR Day 2): Logical inference for argument validation. Constraint Satisfaction (MR Day 1): Resolving conflicts in debates using CSP techniques.	Hybrid Reasoning Systems (RS Day 1): Combines rule-based debates with probabilistic confidence weights. Decision Trees (RS Day 2): Tree-structured reasoning for transparent argumentation.	Tree-of-Thought Reasoning (CGS): Explores multiple reasoning paths (debates) with self-evaluation, mirroring LLM-based cognitive strategies.
	- Generation of Decision Cards with entry/exit ranges - Integration of Kronos forecasting model for quantitative predictions - Position sizing and risk-adjusted strategies	Informed Search (MR Day 1): A* search for optimal strategy generation based on heuristic evaluation. Planning and Optimization (MR Day 1): Synthetic task solving for trade execution.	Generative AI (RS Day 1): Kronos model as a generative component for price forecasting. Reinforcement Learning (RS Day 1): Adaptive strategy refinement based on market feedback.	Generative AI and LLMs (CGS): Kronos as a foundation model for financial time-series prediction, aligned with generative reasoning tasks.

Risk Management Layer	- Multi-perspective evaluation (Optimistic, Neutral, Pessimistic) - Adjustment of position sizing and stop-loss levels - Risk report generation with annotations	Uncertainty Reasoning (MR Day 2): Fuzzy logic for handling imprecise risk thresholds.	Naive Bayes and Bayesian Networks (RS Day 2): Risk assessment using probabilistic models.	Explainable AI (XAI) (CGS): Transparent risk reports emulate human-readable explanations, supporting cognitive collaboration.
		Bayesian Networks (MR Day 2): Probabilistic inference for risk exposure calculation.	Case-Based Reasoning (RS Day 2): Similarity-based risk profiling from historical data.	
Manager Agent & Frontend	- Orchestration of sub-agents - Real-time streaming via WebSocket - Interactive visualization with React interface	Automated Reasoning (MR Day 2): Manager as a central inference engine for task coordination.	Multi-Agent Systems (RS Day 1): Coordinator pattern for agent communication.	Human-AI Interaction (CGS): The React + FastAPI interface enables transparent, traceable reasoning, embodying Cognitive Systems (CGS) principles through intuitive and engaging human–AI collaboration.
		Graph Algorithms (MR Day 1): WebSocket-based data flow modeled as graph traversal.	Real-Time Data Processing (RS Day 1): Async pipelines for scalable data handling.	

12.2.2 Key Insights

- **MR Alignment:** Market Lens employs search and reasoning techniques (e.g., A* search, deductive debates) to ensure systematic problem-solving, mirroring MR's focus on logical inference and optimization.
- **RS Alignment:** The system integrates rule-based, probabilistic, and generative models, showcasing RS's hybrid reasoning approaches for synthetic tasks like strategy generation.
- **CGS Alignment:** By leveraging LLMs for sentiment analysis and Tree-of-Thought-style debates, Market Lens emphasizes cognitive transparency and explainability, core to CGS.

12.3 Installation and User Guide

- (1) **Installation:** Please see the [README documentation](#) for full setup steps.
- (2) **User Guide:** Refer to the System Video for detailed usage instructions.

13 References

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